

A Compact I/Q Network with Center-Tap Coupled Transformer-Based Quadrature Hybrids for Millimeter-Wave Applications

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Abstract—In this paper, a compact broadband in-phase/quadrature (I/Q) network is realized based on transformer-based quadrature hybrids. The coupling capacitors of the proposed hybrid locate at the center taps of the transformer. All output ports of the center-tap coupled quadrature hybrid locate at the same side of the transformer, which makes signal path and power routing much easier. Based on the center-tap coupled hybrid, the proposed I/Q network shows a good performance with less area. A lumped-element model of the center-tap coupled quadrature hybrid is also investigated. Based on the 40-nm CMOS process, an I/Q network with 2-stage transformer-based quadrature hybrids is implemented. Simulation results demonstrate that the proposed structure achieves an image rejection ratio (IRR) of more than 35 dB from 17 GHz to 31 GHz. Furthermore, the area of the proposed structure is 392*379 μm^2 .

Index Terms—in-phase/quadrature (I/Q) network, quadrature generation, millimeter-wave (mm-wave), quadrature hybrid, transformer.

I. INTRODUCTION

Quadrature generation is widely utilized in present radar and wireless communication systems. Quadrature generation circuits are the key components in some structures of phased arrays and transceivers.

To achieve broadband quadrature generation at mm-wave frequencies, poly-phase filters (PPFs) and transformer-based quadrature hybrids are often adopted.

PPFs are sensitive to process and temperature variations. Furthermore, the loss of PPFs increases by 3dB per stage [1]. The application of PPFs is greatly limited by the trade-off between quadrature precision and loss.

The transformer-based quadrature hybrid has been presented for on-chip implementation in [2]. A folded structure of transformer-based quadrature hybrid has been proposed to achieve differential operation in [3]. To improve the quadrature precision and expand the frequency range, an I/Q network consisting of several quadrature hybrids has been presented in [4]. However, it was large size because of several transformers. Moreover, conventional quadrature hybrids put quadrature output ports on the opposite side [5-7]. As a result, signal path and power routing are complicated, which would furtherly increase the area cost. Although some transformer-based quadrature hybrid structures have been proposed to realize the same side quadrature output, the quadrature precision is often degraded, and the theoretical analysis was scarce [8,9].

In this study, a center-tap coupled quadrature hybrid is presented. Based on the center-tap coupled quadrature hybrid, a transformer-based I/Q network is proposed to reduce the area significantly. In Section II, the performance of the 2-stage

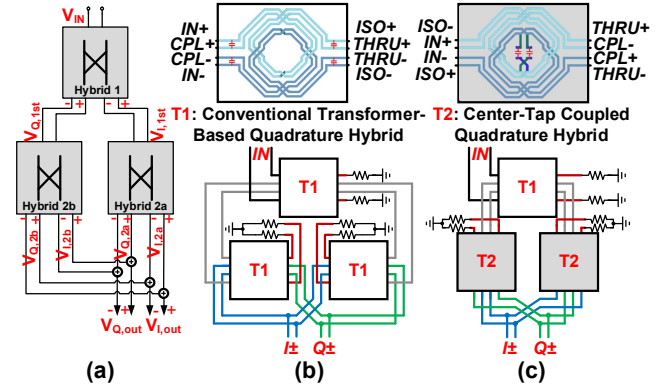


Fig. 1. (a) The structure of the 2-stage transformer-based I/Q network. (b) The layout plan of the conventional 2-stage transformer-based I/Q network. (c) The layout plan of the proposed I/Q network.

transformer-based I/Q network is investigated. The theoretical analysis of the center-tap coupled quadrature hybrid is presented in Section III. The implementation and electromagnetic (EM) simulation results of the proposed network are shown in Section IV. Section V summarizes this paper.

II. ANALYSIS OF THE TRANSFORMER-BASED I/Q NETWORK

The diagram of a two-stage transformer-based I/Q network is shown in Fig. 1(a). The gain of hybrids is defined as

$$A_{I,1st} = \frac{V_{I,1st}}{V_{IN}}, A_{I,2nd} = \frac{V_{I,2nd}}{V_{I,1st}} = \frac{V_{I,2b}}{V_{Q,1st}} \quad (1-2)$$

$$A_{Q,1st} = \frac{V_{Q,1st}}{V_{IN}}, A_{Q,2nd} = \frac{V_{Q,2nd}}{V_{I,1st}} = \frac{V_{Q,2b}}{V_{Q,1st}} \quad (3-4)$$

According to Fig. 1(a), the ratio of output signals can be expressed as

$$\frac{V_{Q,out}}{V_{I,out}} = \frac{A_{I,1st} \cdot A_{Q,2nd} + A_{Q,1st} \cdot A_{I,2nd}}{A_{I,1st} \cdot A_{I,2nd} - A_{Q,1st} \cdot A_{Q,2nd}} \quad (5)$$

Define the image rejection ratio (IRR) as $(V_I - j \cdot V_Q) / (V_I + j \cdot V_Q)$, then the IRR of output signals can be written as

$$|IRR_{out}| = \left| \frac{A_{I,1st} - j A_{Q,1st}}{A_{I,1st} + j A_{Q,1st}} \cdot \frac{A_{I,2nd} - j A_{Q,2nd}}{A_{I,2nd} + j A_{Q,2nd}} \right| \quad (6)$$

Equation (6) can be regarded as $|IRR_{1st} \cdot IRR_{2nd}|$. Thus, the IRR (dB) of a 2-stage transformer-based I/Q network is the sum of the IRR (dB) of each stage.

The output ports of the conventional hybrid (T1 in Fig. 1(b)) locate at the opposite side of the transformer. Based on conventional structures, a typical layout plan is shown in Fig. 1(b). The path is often long. To avoid the coupling between routing lines

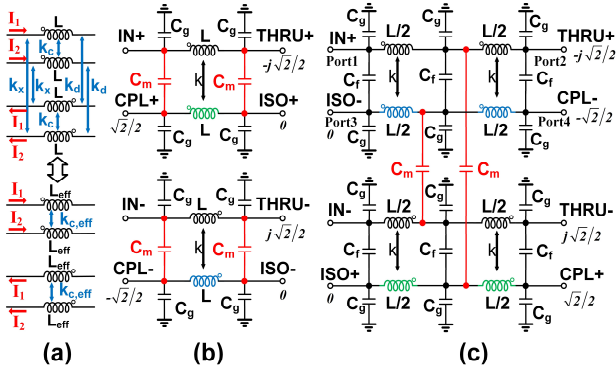


Fig. 2. (a) The coupling effect among four inductors. (b) The lumped element model of the conventional transformer-based quadrature hybrid. (c) The lumped element model of the center-tap coupled quadrature hybrid.

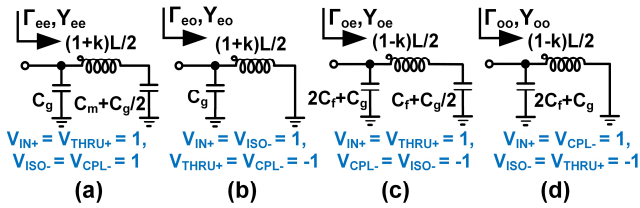


Fig. 3. The equivalent input impedance in (a) even-even mode, (b) even-odd mode, (c) odd-even mode, and (d) odd-odd mode

and transformers, the extra area is required. In addition, the resistors at the isolation ports locate inside transformers, which brings difficulty to the power routing.

The quadrature output ports of the center-tap coupled hybrid (T2 in Fig. 1(c)) locate at the same side of the transformer. With center-tap coupled quadrature hybrids, the length of the path can be reduced and the area can be much smaller, as shown in Fig. 1(c). Moreover, the isolation ports are close to the outside of the overall layout. Thus, the power routing will be simpler.

III. ANALYSIS OF THE CENTER-TAP COUPLED HYBRID

A differential transformer-based quadrature hybrid consists of four coupled inductors, with six coupling coefficients. It can be proved that considering two coupling coefficients is sufficient in a model, as shown in Fig. 2(a). L_{eff} and $k_{c,eff}$ is given by

$$L_{eff} = (1 + k_d)L, k_{c,eff} = \frac{k_c + k_x}{1 + k_d}. \quad (7-8)$$

The schematic of the conventional transformer-based quadrature hybrid is shown in Fig. 2(b). The input port (IN) is driven by a matched source. The network achieves quadrature output at coupled port (CPL) and through port (THRU). The isolation port (ISO) is connected to a matched terminal. In this structure, capacitive coupling (C_m) and inductive coupling (k) are essential. As a result, IN-port and CPL-port are always placed on the same side of the transformer to keep a short distance, while THRU-port and ISO-port are placed on the other side.

The center-tap coupled quadrature hybrid is shown in Fig. 2(c). Based on Fig. 2(b), we can convert a conventional hybrid to a center-tap coupled hybrid in two steps. First, move the coupling capacitors C_m (red ones in Fig. 2(b)) into the center taps

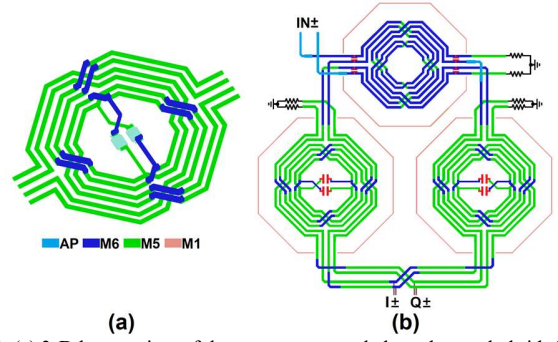


Fig. 4. (a) 3-D layout view of the center-tap coupled quadrature hybrid. (b) The layout of the proposed I/Q network with center-tap coupled transformer-based quadrature hybrids.

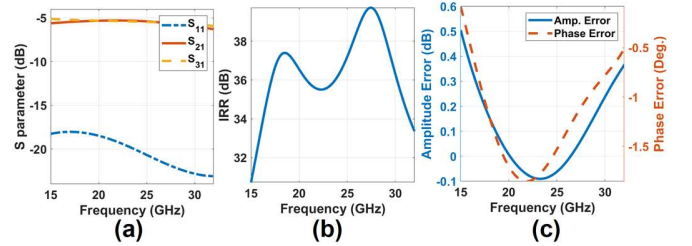


Fig. 5. EM simulation results of the proposed I/Q network: (a) S_{11} , S_{21} , and S_{31} , (c) IRR of the output signals, (d) phase error and amplitude error of the quadrature signals.

of the transformer. Second, swap the green inductors and the blue inductors in Fig. 2(b). The parasitic capacitances of the spiral coils can absorb into C_f and C_g . Parasitic capacitances are often large enough so that no extra capacitors are required.

The center-tap coupled quadrature hybrid can be analyzed in four modes [10]. The input impedance of each port in even-even mode, even-odd mode, odd-even mode, and odd-odd mode can be expressed, referring to Fig. 3.

The S parameters can be calculated by the reflection coefficients in four modes (shown in Fig. 3), which are given as

$$S_{11} = \frac{1}{4}(\Gamma_{ee} + \Gamma_{eo} + \Gamma_{oe} + \Gamma_{oo}) \quad (9)$$

$$S_{21} = \frac{1}{4}(\Gamma_{ee} - \Gamma_{eo} + \Gamma_{oe} - \Gamma_{oo}) \quad (10)$$

$$S_{31} = \frac{1}{4}(\Gamma_{ee} + \Gamma_{eo} - \Gamma_{oe} - \Gamma_{oo}) \quad (11)$$

$$S_{41} = \frac{1}{4}(\Gamma_{ee} - \Gamma_{eo} - \Gamma_{oe} + \Gamma_{oo}). \quad (12)$$

In designing, the target values of S_{11} , S_{21} , S_{31} , and S_{41} are 0, $-j\sqrt{2}/2$, 0, and $-\sqrt{2}/2$, respectively. The reflection coefficients in four modes can be calculated based on (9)-(12). Then the target input admittance in four modes can be expressed as

$$Y_{ee,t} = \frac{j(\sqrt{2}+1)}{Z_0}, Y_{eo,t} = \frac{j(1-\sqrt{2})}{Z_0}, \quad (13-14)$$

$$Y_{oe,t} = \frac{j(\sqrt{2}-1)}{Z_0}, Y_{oo,t} = -\frac{j(\sqrt{2}+1)}{Z_0}. \quad (15-16)$$

Comparing Fig. 3(a)-(d) with (13)-(16), 4 equations can be set up. As an example, assume $\omega_0 = 2\pi \cdot 24$ Grad/s, $Z_0 = 50 \Omega$, and $k = 0.65$, we can solve that $C_f = 7.7$ fF, $C_g = 20.6$ fF, $L = 706$ pH, and $C_m = 50$ fF.

TABLE I
COMPARISON WITH SOME 2-STAGE TRANSFORMER-BASED I/Q NETWORKS AT MM-WAVE FREQUENCIES

	[5] TMTT 2019 (simulation)	[6] TCAS-II 2019 (simulation)	[7] JSSC 2020 (measurement)	[11] TMTT 2021 (simulation)	This Work (simulation)
Process	130nm SiGe BiCMOS	45nm CMOS SOI	45nm CMOS SOI	65nm CMOS	40nm CMOS
Frequency (GHz)	20-35	20-40	25-50	51-66	17-31
Fractional Bandwidth	57%	71%	71%	26%	61%
IRR (dB)	-	-	>35	-	> 35
Phase Error (°)	<1.8	<1.2	<1.8	<2	<1.9
Amplitude Error (dB)	<0.05	<0.3	<0.15	< 0.5	< 0.4
Passive Loss (dB)	2 ¹	1±0.5	2.1 ²	4.5 ³	2.3~3.0
Area (μm ²)	480*450	430*430 ⁴	330*425	560*280 ⁴	392*379
Area / λ ²	0.042*0.040	0.041*0.041	0.039*0.050	0.108*0.054	0.031*0.030

¹ The passive loss at 28 GHz. ² The passive loss at 35 GHz. ³ The passive loss at 60 GHz. ⁴ Estimate based on the chip micrograph

IV. IMPLEMENTATION

Based on the 40nm CMOS process, the layout of the center-tap coupled quadrature hybrid is presented. The 3D layout view is shown in Fig. 4(a). The hybrid is full differential and has a port placement that is convenient for the layout plan. Furthermore, the capacitors are put in the center of the transformer, which won't occupy any additional area or cause any routing inconvenience.

Based on a conventional quadrature hybrid and two center-tap coupled hybrids, a 2-stage I/Q network is implemented. The footprint of the proposed I/Q network is compact, as shown in Fig. 4(b). The area is 392*379 μm².

The S parameters are shown in Fig. 5(a). In the range of 17 GHz to 31 GHz, the passive loss is from 2.3 dB to 3 dB, excluding 3dB power splitting. Fig. 5(b) is the IRR of the output quadrature signal, which is greater than 35 dB. The amplitude error is less than 0.4 dB and the phase error is less than 1.9° from 17 GHz to 31 GHz, shown in Fig. 5(c).

In Table I, we summarize the performance of some 2-stage transformer-based I/Q networks at mm-wave frequencies in recent years. As it can be seen, the proposed I/Q network based on the center-tap coupled quadrature hybrids achieves broadband quadrature generation in a small area.

V. CONCLUSION

This paper presented a center-tap coupled transformer-based quadrature hybrid for mm-wave quadrature generation. The center-tap coupled hybrid had a compact layout and a good port arrangement. Based on the center-tap coupled quadrature hybrid, an area-efficient broadband I/Q network was also proposed. The proposed network is expected to be used in broadband mm-wave systems to significantly reduce the area.

ACKNOWLEDGMENT

This work is supported by the National Key Research and Development Program of China (No. 2020YFB1806304) and

the National Natural Science Foundation of China (Grant No. 61934001).

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